



Enhanced Gateway Card Present REST API

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Enhanced Gateway

Card-Present Transactions

This short document outlines the transaction types, methods and the REST API offered by Enhanced Gateway (EG) to process credit card transactions in a card-present environment.

Transaction Types:

EG REST API provides support for the following transaction types:

The endpoint to receive transaction request is:

{{eg-card-present-endpoint}} = https://postransactions.com/connect/transact

A- Credit Card Transactions

- Sale : *creditsale()*
- Auth: *creditauth()*
- Capture (PostAuth): *creditcapture()*
- Void: *creditvoid()*
- Return: *creditreturn()*
- Adjust: *creditadjust()*
- Tokenize: *gettoken()*

B- Debit Card Transactions

- Debit sale: *debitsale()*
- Debit Return: *debitreturn()*

C- EBT Transactions

- EBT Foodstamp : *foodstampsale()*
- EBT cash: *ebtsale()*
- EBT Return: *ebtreturn()*

D- Gift Card Transactions

- Issue a gift card: *giftissue()*
- Gift Card balance: *giftbalance()*
- Redeem a gift card: *giftredeem()*

E- Batch

- Close batch: *closebatch()*

All card-present transactions use GET calls to perform and send transactions to designated and active devices.

A valid and active EG account is required before you can make any API calls.

For .Net Framework users, we have built an DLL, which you can either directly add to your project, or you can add it from NuGet gallery.

For also provide a JavaScript library for web developments.

Please reach out to our sale engineers' team for more details.

Credit Transactions

Credit sale

A credit sale performs a sale transaction on a qualified, and active device on your account:

Transaction parameters

Parameter	Requirement	Description
method	Required	creditsale()
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
amount	Required	Transaction amount
Token	Conditional – Required for Sale w/Token	A token representation of the card
expirationdate	Optional – Required for sale w/Token	Card's expiration date (mm/yyyy)
tipreq	Optional – Required to request tip at terminal	If set to 1, will ask the customer to select a tip amount
tip	Optional – Required if tip amount is sent along with the transaction amount	If set to an amount, the amount will be added to transaction as the tip amount
merchant	Optional – Required in Multi-Merchant scenarios	Send this parameter with merchant value of 1,...n to process a transaction in a multi-merchant system

Example:

{{eg-card-present-endpoint}}?method=creditsale&account=[Your EG accountNumber]&amount=2.50&ticketid=100&sn=[Your EG active device]

Response:

```
{  
  "TransactionResult": "false",
```

"PTK": "nD5pdVqKNm",
"ResponseCode": "100002",
"ApprovedAmount": "0.00",
"TransactionID": "83654813",
"AuthCode": "000000",
"Token": null,
"CardType": "",
"AccountNum": null,
"ExpDate": null,
"TicketID": "100",
"ResponseMsg": "ABORTED",
"HostCode": null,
"Timestamp": "2022-12-07 18:46:39",
"AvsResponse": null,
"CardHolder": null,
"BusinessId": "xxxxxxxxxx",
"UniqueId": "6390df8f5deda",
"TransType": "",
"EntryMode": null,
"RefNum": "",
"SerialNo": "xxxxxxxxxx",
"AmountDue": null,
"TipAmount": null,
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,

```

"Tvrr": null,
"Aid": null,
"Ts": null,
"Att": null,
"AppLab": null,
"AppPN": null,
"lad": null,
"Arc": null,
"Cid": null,
"Cvm": null,
"BatchNo": "",
"UserId": null,
"Method": "CREDIT"
}

```

After you received the response, first check the value of TransactionResult field. If the value is true, your transaction has been successful, otherwise it has failed. In case of failure, get the value of the ResponseMsg for more details.

To comply with regulations, you need to use the EMV fields, If the chipped card was dipped (or inserted) or contactless, to build your receipt.

```

Terminal: 2290077355
ID: 81622753
Type: SALE
Card Number: 0775 - VISA
Method: CONTACTLESS
AUTH: 792575
Message: OK
AID: A0000000980840
TVR: 0000000000
TSI: 0000
Invoice: 1
Order ID: 1
Customer: CARDHOLDER/VISA

Amount: $0.01
Total: $0.01
Important: Retain this copy for your records

```

Credit Auth

Auth, creditauth, allows you to run a transaction and authorize an amount from the card. An authorization is not final, until you capture the amount in a follow up transaction. Please see Capture.

Transaction parameters

Parameter	Requirement	Description
method	Required	creditauth()
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
amount	Required	Transaction amount
Token	Conditional – Required for Sale w/Token	A token representation of the card
expirationdate	Optional – Required for sale w/Token	Card's expiration date (mm/yyyy)
merchant	Optional – Required in Multi-Merchant scenarios	Send this parameter with merchant value of 1,...n to process a transaction in a multi-merchant system

Example:

```
{{eg-card-present-endpoint}}?method=creditauth&account=xxxxxxxxxx&amount=2.50&ticketid=100&sn=xxxxxxxx
```

Response:

```
{
  "TransactionResult": "true",
  "PTK": "fgrv7BPM4I",
  "ResponseCode": "000000",
  "ApprovedAmount": "0.50",
  "TransactionID": "83657885",
  "AuthCode": "069377",
  "Token": null,
  "CardType": "VISA",
  "AccountNum": "3487",
  "ExpDate": "0225",
  "TicketID": "101",
  "ResponseMsg": "OK",
```

"HostCode": null,
"Timestamp": "2022-12-07 19:06:11",
"AvsResponse": null,
"CardHolder": "JOHN DOE",
"BusinessId": "xxxxxxxxxx",
"Uniqueid": "6390e423321a6",
"TransType": "AUTH",
"EntryMode": "CHIP",
"RefNum": "17",
"SerialNo": "xxxxxxxx",
"AmountDue": "0.00",
"TipAmount": "0.00",
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,
"Tvr": "8000008000",
"Aid": "A0000000031010",
"Tsi": "6800",
"Atc": null,
"AppLab": null,
"AppPN": null,
"Iad": "06061203212000",
"Arc": "Z3",
"Cid": null,
"Cvm": null,
"BatchNo": "1",

```
"UserId": null,  
"Method": "CREDIT"  
}
```

Although a `creditauth()` transaction is very similar to a `creditsale()`, but you have to keep in mind that, an auth is not final until it's captured.

Credit Capture

A `creditcapture()` transaction, accepts the transaction ID of a previously ran `creditauth()` transaction, and attempts to capture the authorized amount, or a lesser or greater amount and finalize the transaction.

Transaction parameters

Parameter	Requirement	Description
method	Required	<code>creditcapture()</code>
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
amount	Required	Transaction ID of the preceding <code>creditauth()</code> transaction.
transactionid	Required	A token representation of the card

Example:

In the below example, we will capture a new amount for the `creditauth()` transaction we performed above.

```
{{eg-card-present-  
endpoint}}?method=creditcapture&account=xxxxxxxx&amount=2.50&ticketid=100&sn=xxxxxxxx&transactionid=83657885
```

Response:

```
{  
  "TransactionResult": "true",  
  "PTK": "Np14SpVL0r",  
  "ResponseCode": "000000",  
  "ApprovedAmount": "0.65",  
  "TransactionID": "83659400",  
  "AuthCode": "000000",  
  "Token": null,  
}
```


"CardType": "VISA",
"AccountNum": "3487",
"ExpDate": "0225",
"TicketID": "100",
"ResponseMsg": "APPROVED OFFLINE",
"HostCode": null,
"Timestamp": "2022-12-07 19:16:08",
"AvsResponse": null,
"CardHolder": null,
"BusinessId": "xxxxxxxxxx",
"Uniqueid": "6390e678b844c",
"TransType": "POSTAUTH",
"EntryMode": "MANUAL",
"RefNum": "18",
"SerialNo": "xxxxxxxxxx",
"AmountDue": "0.00",
"TipAmount": "0.00",
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,
"Tvr": "8000008000",
"Aid": "A0000000031010",
"Tsi": "6800",
"Atc": null,
"AppLab": null,
"AppPN": null,

```
"Iad": "06061203212000",  
"Arc": "Z3",  
"Cid": null,  
"Cvm": null,  
"BatchNo": "1",  
"UserId": null,  
"Method": "CREDIT"  
}
```

Creditcapture() does not require the use to insert a card. It will use the transaction ID of the creditauth() to complete the transaction.

Once you capture the transaction, you can capture it again.

Credit Void

Use this transaction to void (cancel and remove it from the current batch). Use this transaction to void a previously performed creditsale(), creditauth() or creditcapture().

Transaction parameters

Parameter	Requirement	Description
method	Required	Creditvoid()
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
transactionid	Required	A token representation of the card

Example:

```
{{eg-card-present-  
endpoint}}?method=creditvoid&account=xxxxxxxxx&transactionid=83659400&sn=xxxxxxxx&ticketid=100
```

Response:

```
{  
  "TransactionResult": "true",  
  "PTK": "JfNAYzK1eC",  
}
```

"ResponseCode": "000000",
"ApprovedAmount": "0.65",
"TransactionID": "83661085",
"AuthCode": "000000",
"Token": null,
"CardType": "VISA",
"AccountNum": "3487",
"ExpDate": "0225",
"TicketID": "100",
"ResponseMsg": "APPROVED OFFLINE",
"HostCode": null,
"Timestamp": "2022-12-07 19:26:48",
"AvsResponse": null,
"CardHolder": null,
"BusinessId": "xxxxxxxxxx",
"UniquelId": "6390e8f80c2e0",
"TransType": "VOID_POST",
"EntryMode": "MANUAL",
"RefNum": "19",
"SerialNo": "xxxxxxx",
"AmountDue": "0.00",
"TipAmount": "0.00",
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,
"Tvr": "8000008000",

```

"Aid": "A0000000031010",
"Tsi": "6800",
"Atc": null,
"AppLab": null,
"AppPN": null,
"lad": "06061203212000",
"Arc": "Z3",
"Cid": null,
"Cvm": null,
"BatchNo": "1",
"UserId": null,
"Method": "CREDIT"
}

```

Keep in mind that, `creditvoid()` is effective while the batch is still open. Once a batch is closed and transactions have been sent for processing, you cannot use this transaction.

Credit Return

When you need to issue a refund you use `creditreturn()` transaction.

Transaction parameters

Parameter	Requirement	Description
method	Required	Creditreturn()
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
transactionid	Required	Transaction ID of the transaction that you want to Retrun
Amount	Required	Amount to be returned to the card

`Creditreturn()` requires the transaction ID of the target transaction, which you are trying to refund.

`Creditreturn()` requires the physical card to be present or at least card numbers, and card expiration date are available.

Example:

{{eg-card-present-endpoint}}?method=creditreturn&account=xxxxxxxxx&ticketid=100&sn=xxxxxxxx&transactionid=83682494&amount=0.10

**Response:**

```
{
  "TransactionResult": "true",
  "PTK": "Fx5hPVHbeD",
  "ResponseCode": "000000",
  "ApprovedAmount": "0.10",
  "TransactionID": "83684173",
  "AuthCode": "029283",
  "Token": null,
  "CardType": "VISA",
  "AccountNum": "3487",
  "ExpDate": "0225",
  "TicketID": "100",
```

"ResponseMsg": "OK",
"HostCode": null,
"Timestamp": "2022-12-07 21:45:33",
"AvsResponse": null,
"CardHolder": "CARDHOLDER\\VISA",
"BusinessId": "xxxxxxxxxx",
"UniqueId": "6391097da5a28",
"TransType": "RETURN",
"EntryMode": "CONTACTLESS",
"RefNum": "22",
"SerialNo": "xxxxxxxx",
"AmountDue": "0.00",
"TipAmount": "0.00",
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,
"Tvr": "0000000000",
"Aid": "A0000000031010",
"Tsi": "0000",
"Atc": null,
"AppLab": null,
"AppPN": null,
"Iad": "06061203A00000",
"Arc": null,
"Cid": null,
"Cvm": null,
"BatchNo": "1",
"UserId": null,

```
"Method": "CREDIT"
```

```
}
```

If you don't send an amount, you will be asked to enter an amount.

Credit Adjust

We use `creditadjust()` to adjust a previously performed `creditsale()` transaction and add tip to it.

Transaction parameters

Parameter	Requirement	Description
method	Required	Creditadjust()
account	Required	Valid and active EG account number
SN	Required	Active Device serial number
ticketid	Required	A unique ticket ID or invoice number
transactionid	Required	Transaction ID of the sale transaction
tip	Required	Tip amount to be added to the transaction

Example:

```
{{eg-card-present-  
endpoint}}?method=creditadjust&account=1665525990&tip=0.05&ticketid=100&sn=3A904431&transactionid=8369390  
4
```

Response:

```
{  
  "TransactionResult": "false",  
  "PTK": "uscmOeqeq8",  
  "ResponseCode": "100004",  
  "ApprovedAmount": "0.00",  
  "TransactionID": "83694018",  
  "AuthCode": "000000",  
  "Token": null,  
  "CardType": "",
```

"AccountNum": null,
"ExpDate": null,
"TicketID": "100",
"ResponseMsg": "UNSUPPORT TRANS",
"HostCode": null,
"Timestamp": "2022-12-07 22:38:49",
"AvsResponse": null,
"CardHolder": null,
"BusinessId": "xxxxxxxxxx",
"UniquelId": "639115f971935",
"TransType": "",
"EntryMode": null,
"RefNum": "",
"SerialNo": "xxxxxxxx",
"AmountDue": null,
"TipAmount": null,
"SurchargeAmount": "0.00",
"CvResponse": null,
"HostResponse": null,
"ExtraBalance": 0,
"CardBin": null,
"Tc": null,
"Tvr": null,
"Aid": null,
"Tsi": null,
"Atc": null,
"AppLab": null,
"AppPN": null,
"Iad": null,


```
"Arc": null,  
"Cid": null,  
"Cvm": null,  
"BatchNo": "",  
"UserId": null,  
"Method": "CREDIT"  
}
```

Creditadjust() transaction only works if you have a Restaurant, or hospitality setting as your industry setting on your device. In other words, you cannot adjust a sale transaction, if you are in a retail setting.

Tokenize a card

You can call gettoken() transaction to tokenize a card and store it. If your device is properly programmed, and your payment processor has activated tokenization on your account, then each creditsale() transaction, returns a token as well.

Your device must support this feature.

Example:

```
{{eg-card-present-endpoint}}?method=gettoken&account=xxxxxxxx&sn=xxxxxxx
```